qstream

Wire Protocol Specification

Version 0.2 (Draft) — header, manifest request, piece request

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Companion to `SPEC.md` (design & milestones). This document defines the on-the-wire format. Status: Draft — M2 not yet implemented.

1 Introduction

qstream is a peer-to-peer live video streaming protocol carried over plain UDP. One **master** node serves an HLS playlist (manifest + `.ts` segments); **peers** download segments and re-serve them to each other. Every node has a single UDP socket on which it both listens and sends.

UDP gives no ordering, no deduplication and no reliability — everything beyond "send a datagram" is defined here. This document specifies:

- the fixed *datagram header* (section 3),
- the *message catalog* (section 4),
- the *manifest request* happy path (section 5),
- the *piece (segment) request* happy path (section 6).

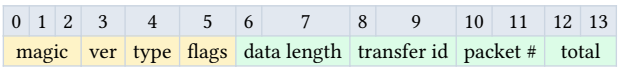
The flow-control and retransmission rules are described in `SPEC.md` §7; only the wire formats are fixed here.

2 Conventions

- All integers are **big-endian** (network byte order).
- Every datagram is exactly one message: fixed header + optional payload.
- Byte offsets are 0-based from the start of the datagram.
- Reserved or unused fields are sent as `0x00` and must be ignored on read.
- Example byte strings are written as space-separated hex, e.g. `51 53 54`.

3 Datagram header

The header is fixed at **14 bytes** and present in every message.



Offset	Size	Field	Description
0	3	magic	`"QST"` (0x51 0x53 0x54). Rejects stray/garbage datagrams.
3	1	version	Protocol version, currently `0x02`.
4	1	message type	One of the codes in section 4.
5	1	flags	ACK type for `ACK` messages (`0x00` progress, `0x04` complete); else `0x00`.
6	2	data length	Payload length in bytes (0..=65535). Datagram = 14 + this.
8	2	transfer id	Correlates all datagrams of one piece transfer; `0x0000` when unused.
10	2	packet number	1-based packet index within a transfer; `0x0000` when unused.
12	2	total packets	Total packets `N` of the transfer; `0x0000` when unused.

Validation. A receiver drops a datagram if the magic or version do not match, or if `data length` exceeds the remaining bytes of the datagram.

3.1 Field usage per message

Fields are only meaningful for certain message types; the others must be zero. A *transfer* is identified by its `transfer id` — chosen at random by the requester and echoed by the responder in every related datagram.

Message	flags	transfer id	packet #	total	payload
HANDSHAKE_REQUEST	—	—	—	—	node name (UTF-8)
HANDSHAKE_RESPONSE	—	—	—	—	node name (UTF-8)
MANIFEST_REQUEST	—	—	—	—	—
MANIFEST_RESPONSE	—	—	—	—	m3u8 contents
SEGMENT_REQUEST	—	✓ transfer	—	—	filename (UTF-8)
SEGMENT_CONTENTS	—	✓ transfer	✓ index	✓ N	file chunk ≤ 1400 B
SEGMENT_NOT_FOUND	—	✓ transfer	—	—	—
ACK	✓ type	✓ transfer	—	—	next range (start, count) — see §6

4 Message catalog

Code	Message	Direction	Status	Payload
`0x01`	HANDSHAKE_REQUEST	peer → master	done (M0)	node name (UTF-8)
`0x02`	HANDSHAKE_RESPONSE	master → peer	done (M0)	node name (UTF-8)
`0x10`	PING	any → any	planned	—
`0x11`	PONG	any → any	planned	—
`0x20`	MANIFEST_REQUEST	peer → master	done (M1)	—
`0x21`	MANIFEST_RESPONSE	master → peer	done (M1)	m3u8 contents
`0x30`	SEGMENT_REQUEST	any → any	spec'd (M2)	filename (UTF-8)
`0x31`	SEGMENT_CONTENTS	any → any	spec'd (M2)	file chunk ≤ 1400 B
`0x32`	SEGMENT_NOT_FOUND	any → any	spec'd (M2)	—
`0x40`	ACK	any → any	spec'd (M2)	next range (start, count)
`0x50`	PEERLIST_REQUEST	peer → master	planned	—
`0x51`	PEERLIST_RESPONSE	master → peer	planned	packed ip:port entries

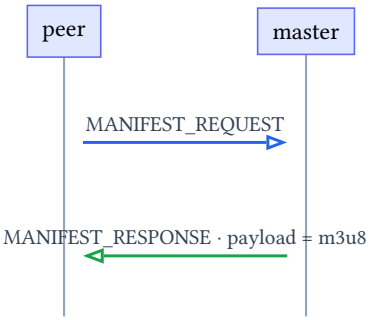
5 Manifest request (happy path)

The peer polls the master for the current HLS playlist. Polling is *stateless*: every request is answered with the latest manifest, there is no session.

5.1 Messages

Message	Format
MANIFEST_REQUEST	`51 53 54` (magic) · `02` (version) · `20` (type) · `00` (flags) · `00 00` (data length) · `00 00` (transfer id) · `00 00` (packet #) · `00 00` (total) — 14 bytes, no payload.
MANIFEST_RESPONSE	Same header with type `21` and `data length` = manifest size; payload = raw m3u8 bytes. Example for the playlist `#EXTM3U\n` (8 bytes): `51 53 54 02 21 00 00 08 00 00 00 00 00 00` then `23 45 58 54 4D 33 55 0A`.

5.2 Exchange



1. The peer sends `MANIFEST_REQUEST` (no payload, no transfer id).
2. The master re-reads its playlist file from disk and replies `MANIFEST_RESPONSE` with the raw contents.
3. The peer writes the response atomically (tmp + rename) to `<data-dir>/live.m3u8`.

Failure handling. If the master cannot read the manifest it replies with an empty payload (the peer keeps its previous copy). A timeout is retried on the next poll — the peer polls every 3 s by default.

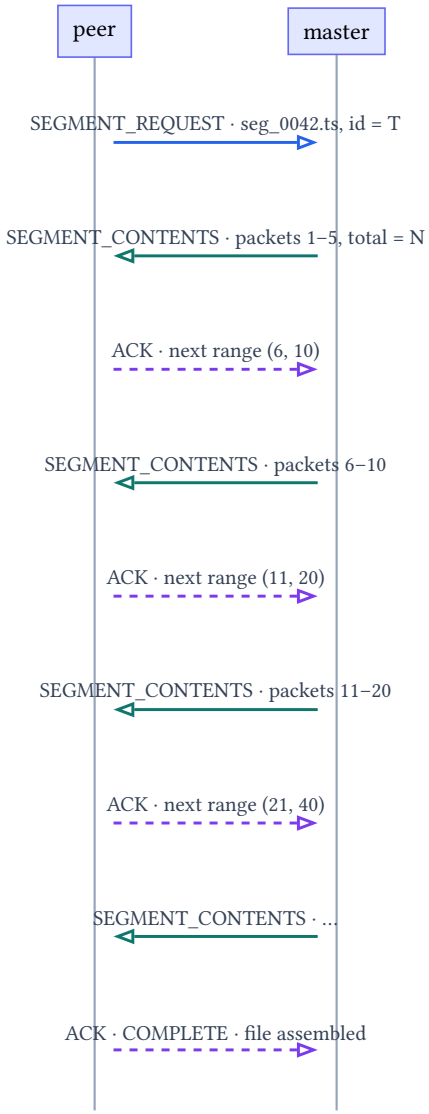
6 Piece (segment) request (happy path)

A piece is one `.ts` segment file, transferred as a sequence of packets. The transfer is **receiver-driven**: the peer explicitly requests the next packet range with every ACK, so there is no sender/receiver window synchronization to lose (see `SPEC.md` §7). `INITIAL_WINDOW = 5`, `MAX_WINDOW = 64`.

6.1 Messages

Message	Format
SEGMENT_REQUEST	Type `30` `transfer id` = fresh random value; payload = filename. Example requesting `seg_0042.ts` with transfer id `0x01A7`: `51 53 54 02 30 00 00 0B 01 A7 00 00 00 00` then `73 65 67 5F 30 30 34 32 2E 74 73`.
SEGMENT_CONTENTS	Type `31` echoes `transfer id`; `packet number` = 1-based index; `total packets` = `N` payload = chunk of the file, ≤ 1400 bytes. First packet of a 60-packet file: `51 53 54 02 31 00 05 78 01 A7 00 01 00 3C` followed by 1400 bytes of file data.
SEGMENT_NOT_FOUND	Type `32` echoes `transfer id`; no payload. The master does not have the requested file.
ACK	Type `40` echoes `transfer id`; payload = next range `(start, count)` as two big-endian u16 (4 bytes). `start` is the first packet wanted, `count` how many. Example: "send me packets 6..10" → `00 06 00 05`. With `flags = 0x04` and an empty payload the transfer is complete.

6.2 Exchange (happy path, window growth 5 → 10 → 20)



1. The peer sends `SEGMENT_REQUEST` for `seg_0042.ts` with a fresh *transfer id* `T`.
2. The master reads the file, computes `N = max(1, ceil(size/1400))` packets, and sends the first window: packets `1..5`.
3. The peer learns `N` from the first content packet and replies `ACK (6, 10)`: the next window, doubled.
4. Windows double on every fully-received range (`5 → 10 → 20 → 40 → …`) up to `MAX_WINDOW = 64`.
5. As soon as the peer has all `N` packets it reassembles the file (out-of-order safe: packets are placed by their `packet number`), writes it atomically, and sends `ACK` with the **COMPLETE** flag.
6. The master frees the transfer state.

Loss handling (short version; full rules in `SPEC.md` §7): if the peer's current range does not fill within the quiet period, it re-sends the same `ACK` (retransmit request). If the master's ack timer fires it re-sends its last range — the peer deduplicates, so blind re-sends are safe. Both sides give up after `8` retries.

7 Appendix: transfer settings

Setting	Value	Meaning
`SEGMENT_PACKET_SIZE`	1400	bytes per content packet
`INITIAL_WINDOW`	5	first requested range size
`MAX_WINDOW`	64	largest range size
`PACE_INTERVAL_MS`	1	sleep between packets (rate limiter)
`WINDOW_QUIET_MS`	150	receiver quiet period before ACKing

`FIRST_RESPONSE_TIMEOUT_MS`	2000	give up if nothing arrives
`WINDOW_RETRY_LIMIT`	8	receiver re-request limit
`ACK_RETRY_LIMIT`	8	sender resend limit
`MAX_CONCURRENT_TRANSFERS`	16	per-node transfer registry bound